

Oğuzhan Oğuz

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Profile

M.Sc. student in Information and Network Engineering at KTH Royal Institute of Technology with a background in electrical and electronics engineering, robotics, computer vision, signal processing, and machine learning. Research experience in LiDAR–visual–inertial SLAM, state estimation, sensor fusion, and robustness under geometrically degenerate conditions, together with hands-on experience in real-time camera–IMU tracking and deep learning. Interested in robot perception, learning-based state estimation, autonomous systems, and machine learning for robotics.

Education

KTH Royal Institute of Technology

Aug 2025 – Present

M.Sc. in Information and Network Engineering

Stockholm, Sweden

Multimedia Processing and Analysis track. GPA: 4.88/5.00.

KTH One-Year Scholarship for Academic Excellence, May 2026.

Middle East Technical University

Aug 2020 – Jun 2025

B.Sc. in Electrical and Electronics Engineering

Ankara, Türkiye

GPA: 3.57/4.00. Ranked 28/396 in class.

Research Experience

KTH Royal Institute of Technology

2026 – Present

Research on Degeneracy-Aware LiDAR–Visual–Inertial SLAM

Stockholm, Sweden

Research conducted with Yixi Cai.

- Investigated the robustness of LiDAR–visual–inertial SLAM in challenging and geometrically degenerate environments using FAST-LIO-VF, focusing on estimator failures under aggressive motion, weak LiDAR geometry, low-texture scenes, visual feature starvation, and perceptual aliasing.
- Designed and evaluated visual innovation gating, sparse information-aware updates, robust Huber weighting, temporal-consistency checks, deterministic ORB feature extraction, degeneracy-aware sensor weighting, and geometric landmark anchoring.
- Built a reproducible ROS-based experimental pipeline and conducted repeated interleaved trials with statistical analysis to distinguish genuine performance improvements from run-to-run stochastic variation.
- Analyzed the limitations of sparse visual features during LiDAR degeneracy, compared FAST-LIO-VF with tightly coupled LiDAR–visual methods including FAST-LIVO2, and investigated recent LiDAR-degeneracy and learned visual-matching approaches toward tightly coupled, degeneracy-aware sensor fusion.

National Magnetic Resonance Research Center

Jul 2024 – Aug 2024

Undergraduate Research Intern

Ankara, Türkiye

- Investigated jitter effects through mathematical modeling and MATLAB simulation under the supervision of Prof. Ergin Atalar.
- Documented methods, simulations, and findings in a technical research report.

Engineering Experience

DEICO Engineering Inc.

Jan 2025 – Feb 2025

Test Engineering Intern

Ankara, Türkiye

- Developed C# interface tools for automated device control and test execution in engineering validation workflows.
- Implemented serial communication routines for reporting, validation, and hardware test automation.

Selected Projects

- **Inside-Out Tracking Sensor Suite** (*B.Sc. Graduation Project*): Developed a real-time inside-out VR tracking system using Raspberry Pi 5, Camera Module 3, and BNO055 IMU for 6-DoF head-pose estimation; implemented camera–IMU fusion, ORB-SLAM3-based position tracking, low-latency wireless transmission, and Python-based 3D visualization. Achieved mean absolute orientation errors of 3.40 degrees (roll), 2.02 degrees (pitch), and 2.93 degrees (yaw), with one-way transmission latency below 6 ms and visualization at approximately 20 FPS.
- **Signal Classification with Deep Learning**: Implemented a 1D residual neural network in PyTorch to classify five analog modulation types from synthetic I/Q data under varying SNR and carrier-frequency offset conditions; developed the complete data-generation, training, and evaluation pipeline.

- **Video Coding with Motion Compensation:** Built and compared intra, conditional-replenishment, and motion-compensated codecs in MATLAB; analyzed bitrate–PSNR trade-offs on QCIF video sequences and compared coding efficiency across motion scenarios.
- **Wiener Filtering for Speech Enhancement:** Designed and compared non-causal, FIR, and causal Wiener filters for speech enhancement using AR-based speech and noise modeling from a single noisy recording; evaluated the methods under a common estimation framework.

Technical Skills

Programming: Python, MATLAB, C#

Machine Learning / Vision: PyTorch, NumPy, SciPy, OpenCV, Matplotlib

Robotics and Perception: ROS, SLAM, LiDAR–Visual–Inertial Fusion, State Estimation, Sensor Fusion, ORB-SLAM3, FAST-LIO-VF

Domains: Computer Vision, Signal Processing, Image and Video Processing, Pattern Recognition, Machine Learning, Digital Communications

Tools: Git, Linux, L^AT_EX, VTK, Siemens NX

Languages

Turkish: Native | **English:** IELTS 7.0 (C1 listening/reading, B2 speaking/writing)